Thruster Characteristics

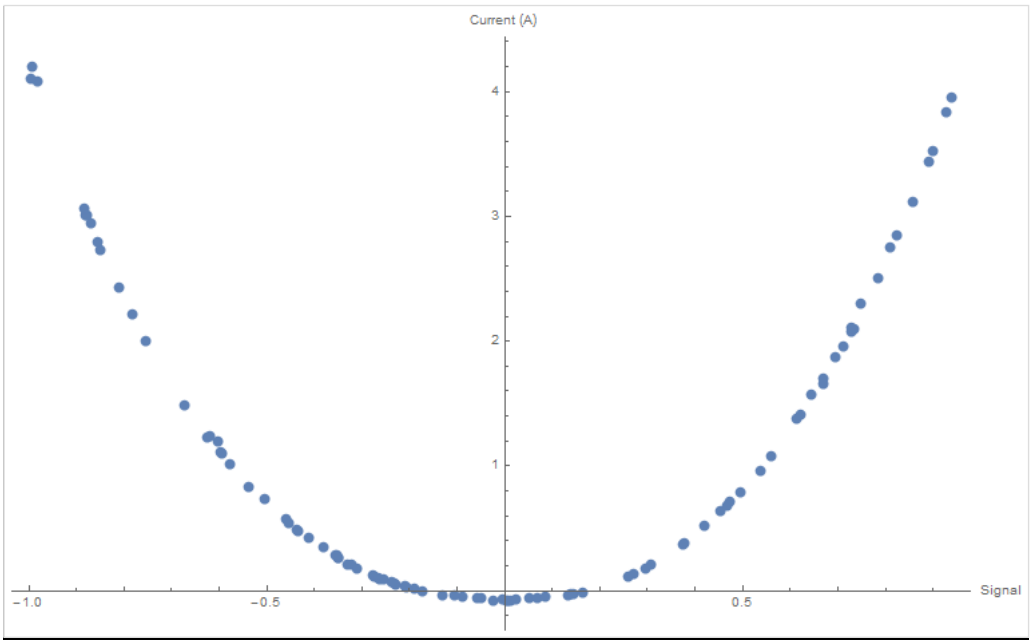
Posted On: **Dec 21, 2015**

Kushal Agarwal

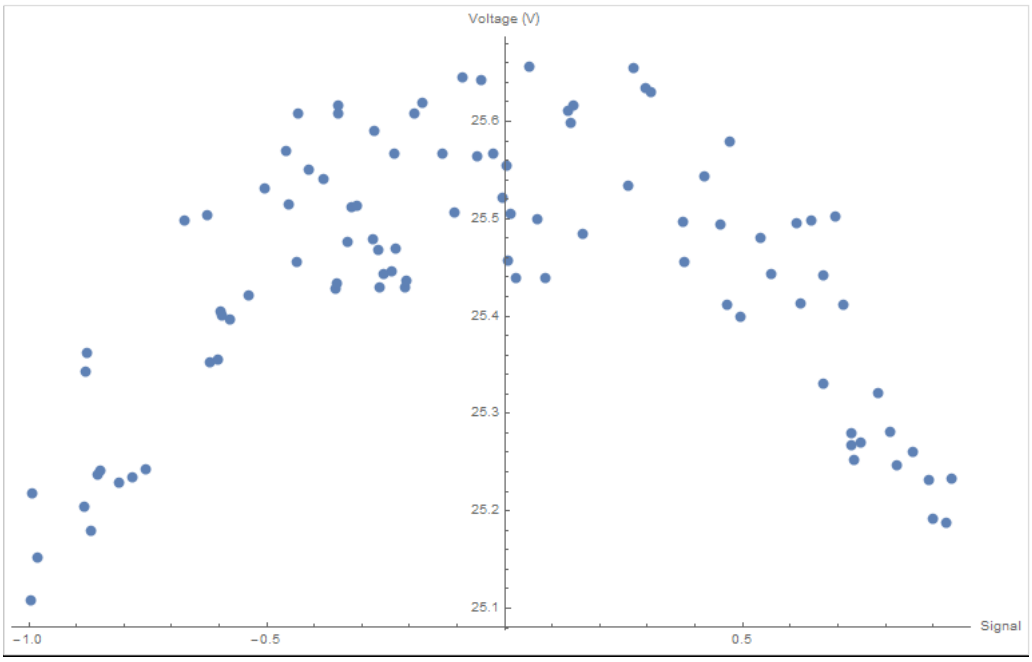
Thanks to everyone who helped with profiling thrusters. Here's a summary of the data we collected.

Electrical Characteristics

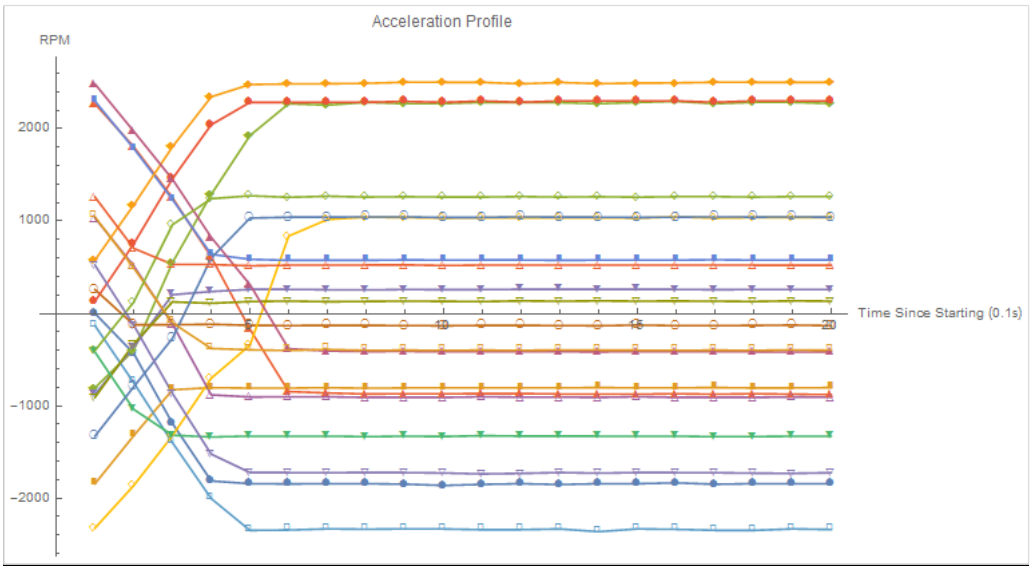
Below is a plot of current per thruster (in amps as a function of signal which goes from -1 to 1): (So a max draw of ~32A for all 7 thrusters)



And also, here is an indication of how well our batteries can source that current: (Looks like 0.5V drop per full-power motor being run)

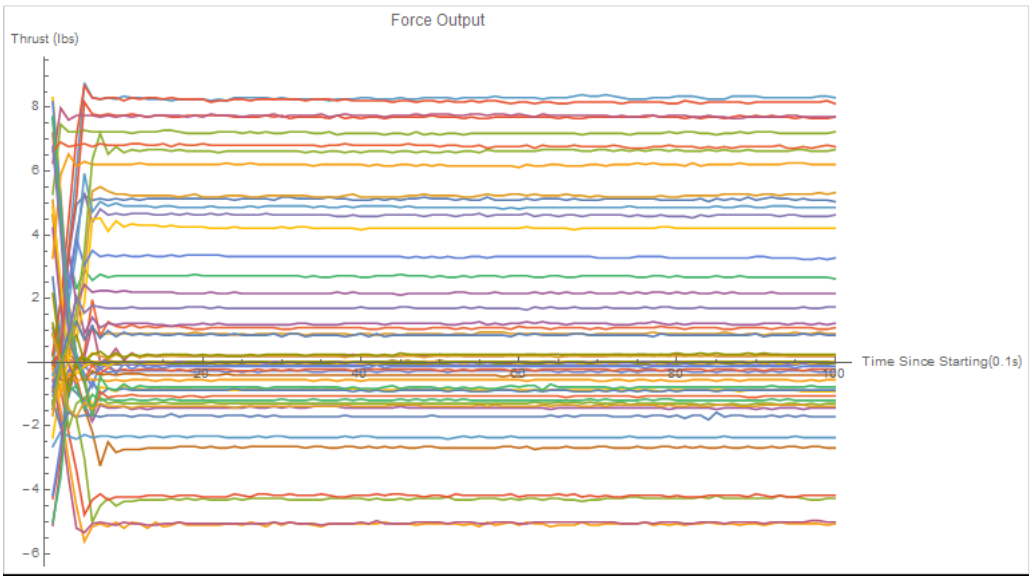


Each line plotted is a 2 second acceleration profile (from some random old speed to a new speed). Note that it takes 0.5-1 sec to completely switch directions and it does so very linearly (so much so that the plot looks kind of like a subway map)



Vortices

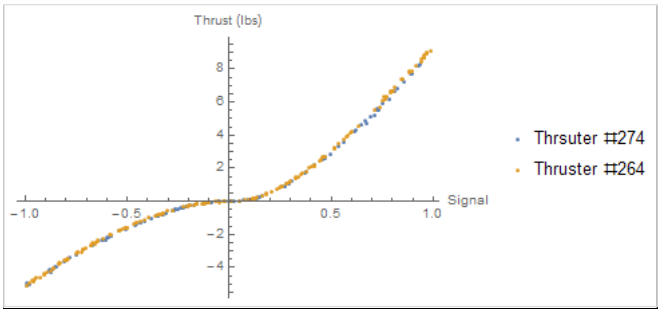
Originally (and based on our test last year in Dabiri's lab), I assumed vortices would form, causing thrust to reduce over time. This turned out to be negligible in the pool. Each line below represents a 10 sec run (from some random old speed to a new speed) -- notice there is no slope to the lines



Since we couldn't create any vortices, we have no reason to use current data in our thrust profile.

Uniformity

Signal vs. Force data from two oppositely rotating thrusters is superimposed. They line up almost perfectly (hence no need to profile all 8 thrusters).



Thruster Profile

To convert force (F) (in units of lbs) into a signal from -1 to 1, I fit the following function (It's kind of arbitrary, just whatever terms helped produced a good fit):

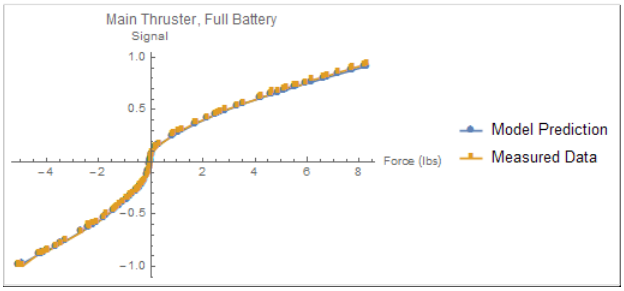
For F positive: signal = [1.72876*F + 2.28643*arctan(15.4867*F) + 5.42665*tanh(0.34581*F)] / [V+0.000294208*V^2]

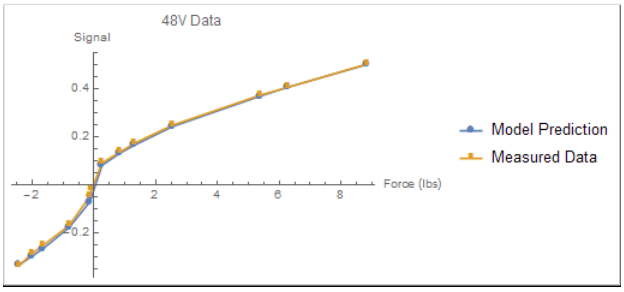
For F negative: signal = [3.06147*F + 2.81235*arctan(10.1538*F) + 5.23219*tanh(0.53369*F)] / [V + 0.000294208*V^2]

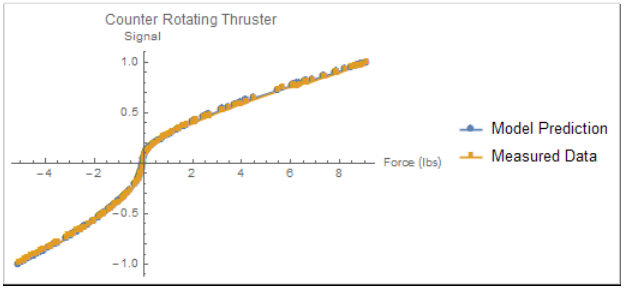
(Where F positive is the direction in which the thruster is more efficient, and V is voltage)

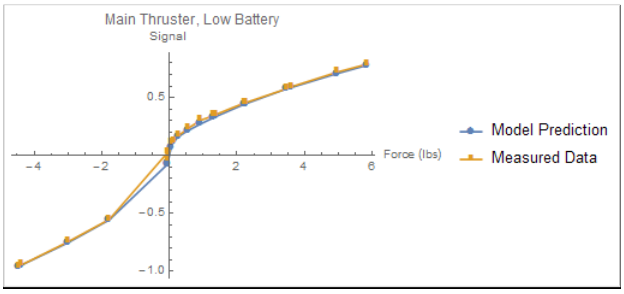
Given this profile, the range of the thruster is -3.5 to 6.5 lbs (At a conservative 20v due to browning out) and -4.8 to 8.8 lbs (at a typical 24v) and -5.3 to 9.7 lbs (at a topped off 25.5v).

Here are some plots showing the fit of this model on the different sets of data we took








Main Thruster, Low Battery
Signal
0.5
−0.5
−1.0
−4
−2
2
4
6
Force (lbs)
Model Prediction
Measured Data